

Jimena Guallar Blasco

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I am a research scientist in natural language processing with a focus on computational linguistics with experience in software engineering. My research interests include the development of reliable high-performance multi-modal AI systems. I have experience developing machine learning tools intended for deployment in safety critical, high- and low-compute environments, such as systems intended for pilot training, which has given me a strong sense of the importance of reliable models aligned with human values.

Education

Johns Hopkins University – MSE in Computer Science (Advisor: Dr. Benjamin van Durme) 2024
Thesis: *“Exploring Vagueness and Ambiguity in Humans and Models Through Visual Question Answering”*
Johns Hopkins University – BS in Computer Science (minor in linguistics) 2024

Experience

Scientist II, Charles River Analytics, a GRVITY Company – Cambridge, MA 2024 – Present

- Lead NLP scientist of five government-funded research programs within the Human Centered AI Division. Lead the development and training of medium sized LLMs and ASR models using 12 GPUs.
- Provided engineering support in the containerization and deployment of natural language systems to edge-devices and government servers using Docker and Kubernetes.
- Applied for and received SBIR research funding totaling over \$900,000
- Lead the Machine Learning reading group to foster a collaborative research culture.

Research Associate, Human Language Technology C.o.E., JHU – Baltimore, MD 2024

- Developed and lead distributed training of multi-modal transformer-based LLMs for high accuracy video-retrieval systems for real-world multilingual videos.
- Organized the development and generation of a large-scale (>200,000 videos) multi-lingual video dataset for for downstream video-retrieval model development.
- Participated in the SCALE program, which focused on generating reliable video-retrieval systems with the collaboration of government intelligence partners. Coordinated the publication of multiple papers from SCALE.

Research Student, Center for Language & Speech Processing, JHU – Baltimore, MD 2020 – 2024

- Studied the divergence of vagueness and ambiguity between LLMs and humans using multi-modal LLMs, and developed methods to mitigate these differences.
- Lead the dataset development of projects studying human linguistic phenomena such as modality (project funded by IARPA), vagueness, and ambiguity.
- Finalist for the Outstanding Undergraduate Researcher Award from the Computing Research Association.

Selected Recent Publications & Invited Talks

For a full publication list, see Google Scholar (link).

1. Invited Talk: *“Transforming Technical Documentation into On-Demand Adaptive Training Content”*, Intersective/Industry Training, Simulation, and Education Conference (I/ITSEC), 2025.
2. R. Kriz, K. Sanders, D. Etter, K. Murray, C. Carpenter, K. Van Ochten, H. Recknor, **J. Guallar-Blasco**, et al., *“MultiVENT 2.0: A Massive Multilingual Benchmark for Event-Centric Video Retrieval”* IEEE CVPR, 2025 (arXiv:2410.11619)
3. E. Stengel-Eskin, **J. Guallar-Blasco**, Y. Zhou, and B. Van Durme, *“Why Did the Chicken Cross the Road? Rephrasing and Analyzing Ambiguous Questions in VQA”*, Proc. 61st ACL, 2023, (arXiv:2211.07516)

Skills

Programming/Markup: Python, Java, LaTeX

Libraries/Frameworks/Tools: PyTorch, NumPy, Matplotlib, Pandas, OpenCV, NLTK, Scikit-Learn, Amazon MechanicalTurk, HuggingFace Transformers, Django, Pillow, Keras

Systems/Infrastructure: Docker, CUDA, Kubernetes